REFERENCE TRANSLATION prepared by Van Phong Technical and Trading Co., Ltd. The published original is the authoritative text; see it for figures.

Influence of Cold Treatment on the Mechanical Properties and Wear Resistance of Hydraulic Breaker Pick Materials

Le Thanh Binh, V.I. Bolobov, Nguyen Khac Linh

Gornyy informatsionno-analiticheskiy byulleten' (Mining Informational and Analytical Bulletin, scientific and technical journal), 2017, No. 12 (Special Issue 25), pp. 3–11 · Moscow: Gornaya Kniga Publishing House · ISSN 0236-1493 · UDC 622.23.05 · DOI: 10.25018/0236-1493-2017-12-25-3-11

Abstract

It is shown that cold treatment (at −75 °C) of hydraulic breaker pick materials – steels 38XM, У8 and X12МФА – leads to a decrease (by 44–82%) in the content of retained austenite and an increase (by 26–99%) in the content of carbides in hardened steel, which is accompanied by an increase in hardness (by 1.4–2.1%) and abrasive wear resistance (by 10–31%) of the tested materials, with a simultaneous decrease in impact toughness (by 19–24%). Using steel X12МФА as an example, it is shown that cryogenic treatment (at −196 °C) followed by low-temperature tempering of the high-carbon high-alloy steels used for manufacturing hydraulic breaker picks leads to a significant increase in their wear resistance (up to 98%) and impact toughness (up to 32%).

Keywords: *hydraulic breaker picks, carbon and alloy steels, cold treatment, cryogenic treatment, increase in wear resistance.*

Introduction

The pick is the replaceable part of a hydraulic breaker that interacts directly with the medium being broken. During operation of the breaker, especially on hard and abrasive rocks, the pick is subjected to considerable stresses, impact loads and intensive abrasive wear [1] (Fig. 1). For this reason, the pick must be made of a material with increased strength, hardness and wear resistance combined with sufficient resistance to impact loads. Medium- and high-carbon steels with various degrees of alloying are usually used for this purpose.

[Figure 1 — see original PDF, page 4] Appearance of a JCB HM380 hydraulic breaker pick after service

The main technological operations in pick production are forging, machining and heat treatment. Typical heat treatment (THT) includes austenitizing of the steel used, oil quenching and low-temperature tempering. The structure of the steel after heat treatment is martensite with a certain amount of retained austenite and carbides. The retention of austenite in the structure of quenched steel is due to the martensite finish temperature M_f being lower than the temperature of the quenching medium used. The value of M_f and the amount of retained austenite depend on the chemical composition of the steel and the heat treatment parameters: the higher the carbon content of the steel, the lower the M_f value and the greater the content of retained austenite [2–4].

The presence of "soft" austenite in the structure of the pick material reduces its hardness, strength and wear resistance. On the other hand, under the action of the large impact loads acting on the pick during operation of the hydraulic breaker, part of the retained austenite of the steel transforms into martensite while the pick is in service. Since the specific volume of martensite is 4.3% greater than that of austenite, this transformation leads to distortion of the pick dimensions, which affects the trajectory of the hammer's motion.

In order to reduce the content of retained austenite in quenched steels, cold treatment or cryogenic treatment (at the temperature of liquid nitrogen, −196 °C) can be used as an additional heat-treatment operation, in which the product is cooled to a temperature below the M_f of the steel used and held at this temperature for a long time. Since the thermal stability of retained austenite in quenched steel increases during tempering [5], cold treatment is performed immediately after quenching (followed by tempering).

Another important reason for the beneficial effect of cold is associated with the residual stresses arising in products subjected to machining. As the level of stresses arising from machining increases, the density of defects in the crystal structure of materials – voids, stacking dislocations, etc. – increases, which leads to an increase in the interatomic distance. When this distance exceeds a certain critical value, cracks appear and the product fails [6].

The third law of thermodynamics states that entropy tends to zero at absolute zero temperature. This principle is used to relieve residual stresses in metallic materials by cold treatment. Prolonged exposure to low temperatures helps bring the system into a state of equilibrium, resulting in a decrease in the density of defects in the material and a reduction in the interatomic distance in the crystal lattice. Densification of the crystal structure leads to a significant improvement in the wear resistance of the material and also increases its corrosion resistance and fatigue strength [7].

Moreover, cold treatment leads to the precipitation from martensite of a large number of very fine carbide particles, called η-carbide. According to [5], this occurs due to the contraction of the martensite crystal lattice as the temperature decreases and its transition to a more unstable thermodynamic state.

For this reason, cold treatment is widely used as an additional operation after quenching to improve the wear resistance of tool steels [2, 8–16]. The aim of this work was to study the influence of cold treatment and cryogenic treatment on the structure, mechanical properties and abrasive wear resistance of steels (38ХМ, У8 and Х12МФА) subjected to typical heat treatment and commonly used as hydraulic breaker pick materials.

Experimental Procedure

Steel specimens, immediately after oil quenching (from 870 °C for steels 38ХМ and Х12МФА and from 800 °C for У8), were placed in a cryostat with a coolant (T = −75 °C) or with liquid nitrogen (T = −196 °C), where they were held for 5 and 72 hours, respectively, and then tempered at 200 °C.

The content of carbides and retained austenite in the structure of the steels (with a sensitivity threshold of 0.1%) was determined using an Axioplan-2 microscope and KS 300 3.0 software. The KCV and HV values were determined on a Zwick/Roell RKP450 pendulum impact tester and a Zwick/Roell ZHU universal hardness tester.

Abrasive wear tests were carried out according to the method of [17, 18] on the rig whose diagram is shown in Fig. 2, at a machine spindle speed of 7.5 s^{-1}. Granite, as the rock most frequently broken by hydraulic breakers, was used as the abrasive medium 1 (Fig. 2). Under a constant static load P (100 N), the end face of specimen 2 (cylinder, D ≈ 8 mm) of steel subjected to one or another heat

treatment was worn for 50 min, with its mass m measured (weighing accuracy 1.0 mg) every 10 min and its loss Δm calculated. After each measurement the plate of abrasive material was shifted relative to the specimen, which ensured that in each cycle the metal was in contact with a fresh abrasive surface.

[Figure 2 — see original PDF, page 6] Diagram of the rig for abrasive wear testing of specimens: 1 – plate of abrasive material; 2 – test specimen; 3 – holder; 4 – spindle of a vertical drilling machine; 5 – water supply device

From the experimental results, for each test series a graph was plotted of the cumulative mass loss ΣΔm of the specimen versus the duration t of abrasive exposure. An extrapolation straight line was drawn through the experimental points; the slope (tangent of the angle of inclination) of this line was taken as the wear rate K, and its reciprocal as the wear resistance I of the material.

Experimental Results

The results of the microstructural studies are presented in Table 1.

As follows from the data in Table 1, cold treatment leads to a substantial decrease in the content of retained austenite in high-carbon steels У8 (~0.8% C) and Х12МФА (~1.5% C), in which it was retained in significant amounts after quenching. It also leads to an increase in the content of carbides in the quenched alloy steels (38ХМ, Х12МФА). Moreover, the lower the cold-treatment temperature, the higher its effectiveness.

As the results of the abrasive wear experiments showed, for all tested steel specimens the wear process is described by a linear dependence (Fig. 3):

$$\Sigma\Delta m = K \cdot t,$$

where K is the wear rate, which depends on the steel grade and the type of heat treatment.

Table 1. Influence of cold treatment on the amount of retained austenite and carbides in the structure of the steels

Steel grade	% retained austenite			% carbides		
	no cold treatment	cold at −75 °C	cold at −196 °C	no cold treatment	cold at −75 °C	cold at −196 °C
38ХМ	≤0.10	≤0.10	≤0.10	0.17	0.33 (+98.9%)	0.58 (+243.2%)
У8	2.21	1.24 (−43.8%)	0.84 (−61.9%)	≤0.10	≤0.10	≤0.10
Х12МФА	15.65	2.78 (−82.3%)	0.50 (−96.8%)	5.41	6.80 (+25.7%)	10.84 (+100.4%)

The determined values of wear resistance (I = 1/K), impact toughness and hardness of the steels, as the arithmetic mean of the results of 3 experiments, are presented in Table 2 and Fig. 4.

Table 2. Results of physical-mechanical tests and wear experiments on steels subjected to typical heat treatment (THT) and to treatment with cold

Steel grade	Type of treatment	Parameter value		
		KCV, J/cm^2	HV	I, min/mg
38ХМ	THT	40.6	529	0.50

Table 2. Results of physical-mechanical tests and wear experiments on steels subjected to typical heat treatment (THT) and to treatment with cold				
Steel grade	**Type of treatment**	**Parameter value**		
		KCV, J/cm^2	**HV**	**I, min/mg**
	THT + cold (−75 °C)	33.0 (−18.7%)	536 (+1.4%)	0.61 (+23.2%)
	THT + cold (−196 °C)	35.0 (−13.8%)	539 (+2.0%)	0.67 (+34.7%)
У8	THT	5.1	707	0.63
	THT + cold (−75 °C)	3.9 (−23.5%)	717 (+1.4%)	0.68 (+9.6%)
	THT + cold (−196 °C)	5.1 (+0.0%)	730 (+3.2%)	0.76 (+21.2%)
Х12МФА	THT	3.1	694	1.12
	THT + cold (−75 °C)	2.5 (−19.4%)	708 (+2.1%)	1.47 (+30.9%)
	THT + cold (−196 °C)	4.1 (+32.3%)	750 (+8.2%)	2.22 (+97.8%)

As can be seen from Table 2 and Fig. 4, as the treatment temperature decreases, the hardness of all steel grades increases, most noticeably for steel Х12МФА (by 8.2% at −75 °C). The same material also shows the greatest increase in wear resistance (by 97.8%). The impact toughness of all materials decreases as a result of treatment at −75 °C, but increases at −196 °C, exceeding for steel Х12МФА the KCV value after typical heat treatment (4.1 compared with 3.1 J/cm^2).

[Figure 4 (numbered “Fig. 1” in the printed original) — see original PDF, page 8] Influence of cold-treatment temperature on the mechanical properties and wear resistance of hydraulic breaker pick materials – steels 38ХМ (1), У8 (2) and Х12МФА (3) (panels: HV, KCV and I, min/mg versus treatment temperature, °C)

Discussion of Results

The difference in the degree of influence of cold treatment on the wear resistance of the steels is explained by the fact that this characteristic depends not only on the hardness of the steel, but also on the type, volume and shape of the carbides present in it.

Steel Х12МФА shows the most noticeable improvement in hardness and wear resistance because after quenching it contains a significant amount of retained austenite (more than 15%), which almost completely transforms into martensite after cold treatment at −196 °C. In addition, it has the highest content of carbon (~1.5%) and alloying elements (~13%) of the steels analysed, which ensures, after cold treatment, an increase of up to 10.8% in the amount of fine carbide particles.

Steel 38ХМ does not show such a noticeable improvement in hardness (only by 2.0%) and wear resistance (by 34.7%) because of the practical absence of retained austenite in its structure after quenching. The increase in HV and I as a result of treatment is due solely to the increase in the amount of fine carbide particles. Even so, the 1.35-fold increase in wear resistance is also significant.

For steel У8, on the contrary, the increase in hardness (by 3.2%) and wear resistance (by 21.2%) as a result of treatment is associated with the transformation of the retained austenite in the steel structure into martensite.

Cold treatment at −75 °C reduces the impact toughness of all steels to approximately the same extent (from 18.7 to 23.5%). At the same time, lowering the treatment temperature to −196 °C leads, for all steels, to an increase in KCV compared with −75 °C; for У8 the impact toughness reaches the KCV value of the initial quenched steel, and for steel Х12МФА it even exceeds it (by 32.3%). The improvement in impact toughness as a result of cryogenic treatment can be explained by the uniform

spatial distribution of the precipitated homogeneous fine carbides [19, 20], or it may be associated with microstructural changes occurring within the martensite itself [13].

From the results presented it can be concluded that cold treatment may be regarded as a promising additional heat-treatment operation for hydraulic breaker picks to increase their wear resistance. The most preferable treatment temperature should be considered −196 °C, at which, along with an increase in the hardness and wear resistance of the pick material, its impact toughness also increases.

Conclusions

1. As a result of cold treatment at −75 °C, most of the retained austenite in the structure of the quenched high-carbon steels У8 and Х12МФА used for manufacturing hydraulic breaker picks transforms into martensite, which leads to an increase in the hardness and wear resistance of the steel, as well as to a slight decrease in its toughness.
2. Using steel Х12МФА as an example, it is shown that cryogenic treatment (at −196 °C) followed by low-temperature tempering of the high-carbon high-alloy steels used for manufacturing hydraulic breaker picks leads to a significant increase in wear resistance (up to 98%) and impact toughness (by 32%). The positive effect of the treatment can be explained by the precipitation of a large number of fine carbide particles uniformly distributed in the steel structure.
3. Supplementing the typical heat treatment of hydraulic breaker picks with cryogenic treatment at −196 °C (between the quenching and low-temperature tempering operations) will help increase the hardness, wear resistance and impact toughness of the steel used.

References

1. Болобов В. И. и др. ...ным ударом // Горное оборудование и электромеханика. [Title partly illegible in the scanned original; per the paper's English reference list: Bolobov V. I. i dr. Gornoe oborudovanie i jelektromehanika, 2017, no 6, pp. 43-48.]
2. Клименко А. П. и др. Холод в машиностроении. – М.: Машиностроение, 1969. – 248 с.
3. Чудаков Е. А. Машиностроение. Энциклопедический справочник. Раздел 2 Материалы машиностроения. Том 3. М.: ГОНТИ, 1947. – 341 с.
4. Новиков И. И. Теория термической обработки металлов. – М.: Металлургия, 1978. – 392 с.
5. Yun Dong и др. Deep Cryogenic Treatment of High-speed Steel and Its Mechanism // Heat Treatment of Metals. – 1998. – №. 3. – pp. 55–59.
6. Patil P. I. и Tated R. G. Comparison of Effects of Cryogenic Treatment on Different Types of Steels: A Review // International Conference in Computational Intelligence (ICCIA) – Proceedings published in International Journal of Computer Applications® (IJCA). – 2012. – pp. 10–29.
7. Tated R. G. и др. Improvement in Tool Life of Cutting Tool by Application of Deep Cryogenic Treatment // 7th International Tooling Conference Held, Politecnico di Torino, Itlay. – 2006. – pp. 135–141.
8. Collins D. N. Deep Cryogenic Treatment of Tool Steels: A Review // Heat Treatment of Metals. – 1996. – pp. 40–42.
9. Sub-zero Treatment of Steels – Technology/ Processes/ Equipment. Linde Gas, 2010. 20 p.
10. ASM Handbook. Том 4 Heat Treating. ASM International, 2002. 2173 p.

11. Candane D. и др. Effect of Cryogenic Treatment on Microstructure and Wear Characteristics of AISI M35 HSS // International Journal of Materials Science and Applications. – 2013. – Vol. 2. – pp. 56–65.
12. Collins D. N. Cryogenic Treatment of Tool Steels // Advanced Materials and Processes. – 1998. – pp. 23–29.
13. Collins D. N. и Dormer J. Deep Cryogenic Treatment of a D2 Cold-work Tool Steel // Heat Treatment of Metals. – 1997. – pp. 71–74.
14. Kalsi N. S. и др. Cryogenic Treatment of Tool Materials: A Review // Materials and Manufacturing Processes. – 2012. – Vol. 25. – pp. 1077–1100.
15. Paulin P. Cold Cuts // Cutting Tool Engineering. – 1992. – pp. 61–66.
16. Yugandhar T. и Krishnan P. K. Cryogenic Treatment and It's Effect on Tool Steel // 6th International Tooling Conference, Karlstad University. – 2002. – pp. 671–684.
17. Болобов В. И. и др. Износостойкость стали 110Г13Л в различных абразивных средах // Записки горного института. – 2014. – Т. 209. – С. 17–22.
18. Болобов В. И. и др. Износостойкость стали Гадфильда при больших удельных нагрузках // Горное оборудование и электромеханика. – 2012. – № 10. – С. 12–14.
19. Bensely A. и др. Influence of Deep Cryogenic Treatment on Alloy Carbide Precipitations and Mechanical Properties of AISI M2 High Speed Tool Steel // Cold Facts. – 2012. – 28 (2). – С. 4–9.
20. Das Debdulal и др. Influence of Sub-zero Treatments on Fracture Toughness of AISI D2 Steel // Materials Science and Engineering: A – 2010. – Vol. 528. – pp. 589–603.